

AJMAL KHAN

Software Engineer | Kubernetes Administrator

+923334483486 | ak1489007@gmail.com | linkedin | Peshawar, Pak | Github



SKILLS

With a half-decade of IT experience, I have mastered the end-to-end engineering flow: from development of software and deployment through DevOps practices on cloud or on-premises. This three-step progression reflects my expertise in building, deploying, and scaling applications seamlessly across environments.

Technical Proficiency:

- Development:** JavaScript | React | React Redux | Python | Django | Django REST Framework | Django-ORM | WSGI (Gunicorn) | PostgreSQL | pgBackRest | Web Server (Nginx) | JWT | OAuth | Celery | Redis | Git | GitHub
- DevOps:** Linux | Docker | Kubernetes | Terraform | Jenkins | CI/CD | Bash Scripting | Scripting with Python | Prometheus | Grafana | Kubernetes Networking (CNI Plugins) | Helm | Kubectl | Containerd
- Cloud:** AWS (IAM | EC2 | EKS | VPC | S3 | RDS | Fargate (Serverless) | CloudWatch | ACM))

WORK EXPERIENCE

Role: DEVOPS ENGINEER (Remote)

2023/June-2025/Jan

Company: **Pellow Software House**

- Designed and deployed a production-grade Kubernetes cluster using kubectl, containerd with worker nodes across regions for fault tolerance and redundancy.**
- Prior to deployment and infrastructure provisioning, the Kubernetes (k8s) cluster was created using kubectl as the core tool, containerd as the container runtime, and kubelet as the service. Other essential k8s components were provisioned using static pods, configured in the `/etc/kubernetes/manifests` directory.
- Secured k8s components with self-signed and CA-signed certificates, and established secure SSH for cluster management.
- Automated CI/CD pipelines using Jenkins to extract application code from GitHub, containerize microservices (authentication, payments, scheduling, etc.) with Docker, to ensure high availability.
- Provisioned infrastructure using Terraform (IaC) and deployed the multi-tier application on AWS, ensuring zero downtime and seamless integration.
- Conducted stress testing and containerized dependencies to maintain uninterrupted application performance.

Role: REST-APIS DEVELOPMENT AND DEVOPS ENGINEER

2022/Dec-2023/June

Company: **AntonX, Peshawar, Pakistan**

- Provisioned AWS infrastructure using Terraform and deployed a multi-tier application on AWS EKS with a microservices architecture, integrated hybrid nodes from on-premises infrastructure.**
- Developed secure REST APIs using Python's Django REST Framework, integrated with Celery and Redis for asynchronous task processing and scalability.
- Designed a secure VPC with public/private subnets, leveraging AWS CloudWatch, NAT/Internet Gateways, ACM, IAM, and EC2 bastion hosts for robust and secure deployments.
- Utilized AWS EKS managed control plane and worker nodes to deploy containerized applications with restricted network policies for enhanced security.
- Configured Application Load Balancer (ALB) using AWS Load Balancer Controller (LBC) for efficient traffic routing and scalability.
- Monitored and troubleshooted Kubernetes clusters and applications using Prometheus, Grafana, and DaemonSets, deploying Prometheus within the cluster to reduce costs and centralize monitoring.

Role: SOFTWARE DEVELOPER (INTERNEE)

2022/Mar-2022/Oct

Company: **University Of Peshawar, Peshawar**

- Developed and deployed a full-stack monolithic application on a Debian-based Linux server using JavaScript, Python, and PostgreSQL.**
- Designed a decoupled architecture with React frontend and Django REST Framework backend, integrated via CORS, and deployed as individual services on Ubuntu.
- Built dynamic UIs with React, Redux, Yup, Axios, and Recharts, and implemented secure REST APIs with JWT/OAuth (Google) authentication.
- Optimized SQL queries to ensure efficient database design and management.
- Utilized PostgreSQL database performance, managed application files and directories following Linux filesystem hierarchy (FHS) standard, and configured backups using pgBackRest for disaster recovery.

EDUCATION

BS IN COMPUTER SCIENCE

2023/Mar

From: **University Of Peshawar, Peshawar, Pakistan**